MATHS MANTHAN
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 07** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 10: Circles** **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS (LA) questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal choices.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. From a point P, two tangents PA and PB are drawn to a circle with centre O. If $\angle APB = 50°$, then $\angle AOB$ is: [1]

(A) $100°$ (B) $130°$ (C) $140°$ (D) $150°$

Q2. In a right triangle ABC right-angled at C, $BC = 5$ cm and $AC = 12$ cm. The inradius r of the incircle is: [1]

(A) 1.5 cm (B) 2 cm (C) 2.5 cm (D) 3 cm

Q3. If chord AB of length 16 cm of a circle of radius 10 cm is tangent to an inner concentric circle, then the radius of the inner circle is: [1]

(A) 5 cm (B) 6 cm (C) 7 cm (D) 8 cm

Q4. If tangents PA and PB are drawn to a circle with centre O such that $\angle APB = 120°$, then the relation between OP and tangent AP is: [1]

(A) $OP = AP$ (B) $OP = 2AP$ (C) $OP = \sqrt{3}AP$ (D) $AP = 2OP$

Q5. A line that intersects a circle in two distinct points is called a: [1]

(A) Tangent (B) Secant (C) Radius (D) Normal

Q6. **Assertion (A):** If two tangents TP and TQ are drawn from an external point T to a circle with centre O, then $\angle PTQ = 2\angle OPQ$.
Reason (R): In $\triangle TPQ$, $TP = TQ$ implies $\angle TPQ = \angle TQP = \frac{1}{2}(180° - \angle PTQ)$, and $\angle OPT = 90°$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. A chord of length 16 cm of a circle of radius 10 cm is tangent to an inner concentric circle. Find the radius of the inner circle. [2]

Q8. If PA and PB are two tangents drawn from an external point P to a circle with centre O such that $\angle APB = 60°$ and $PA = 7$ cm, find the length of chord AB. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. Two concentric circles are of radii R and r $(R > r)$. A chord AB of the outer circle touches the inner circle at P. Prove that the length of chord AB is $2\sqrt{R^2 - r^2}$. If $R = 13$ cm and chord length is 24 cm, find r. [3]

Q10. In an isosceles $\triangle ABC$ with $AB = AC = 10$ cm and $BC = 12$ cm, a circle is inscribed. Find the lengths of the tangent segments into which the sides are divided by the points of contact. [3]

SECTION D: Long Answer / HOTS Type (2 × 5 = 10 Marks)

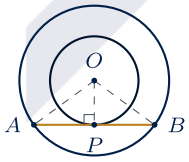
Q11. A triangle ABC is drawn to circumscribe a circle of radius 4 cm such that the segments BD and DC into which BC is divided by the point of contact D are of lengths 8 cm and 6 cm respectively. Find the sides AB and AC. [5]

Q12. Prove that the opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle. Using this result, if $\angle AOB = 125°$ where O is the centre, find the measure of $\angle COD$. [5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Concentric Circular Running Track and Direct Sprint Chords

A sports stadium features two concentric circular tracks of inner radius $r = 20$ m and outer radius $R = 25$ m sharing a common centre O. A straight sprint corridor AB is painted along a chord of the outer track such that it just grazes (is tangent to) the inner track at point P.



(i) State the angle between radius OP and the sprint chord AB. [1]

(ii) Find the length AP of the half-chord. [1]

(iii) Calculate the total length of the sprint corridor AB, OR find the area of the track region between the concentric boundaries. [2]

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